Evolution in real time

Richard Lenski, a professor at Michigan State University, established the Long-term Experimental Evolution project in 1988. For more than 25 years, he studied 59,000 generations of the *E. coli* bacterium. During this time, he observed that the species used the glucose solution it lived in more efficiently, increasing in size but also growing faster. Also, a new species had evolved that was able to use a compound in the solution called citrate, which the parent bacterium could not. Evolving bacteria can pose a potential threat to humans. Increasing antibiotic use destroys many disease-causing bacteria, but not those with mutations that make them resistant to the drugs. As the non-resistant bacteria are killed off, the resistant strains become more dominant, multiplying and passing on their mutations to future generations. That is natural selection at work.